

Build Your Own LLM with Open Source Frameworks and Tools

Large language models are machine learning models designed for a range of language-related tasks such as text generation and translation. Here's how open source software can help you build your own large language model.



A large language model (LLM) is an advanced artificial intelligence system designed to process and generate human-like text based on vast amounts of data. These models leverage deep learning techniques, particularly transformer architectures, to understand, predict, and generate natural language.

Popular examples include:

- OpenAI's GPT
- Google's Gemini
- Meta's Llama
- DeepSeek

LLMs can perform various tasks such as text generation, summarisation, translation, sentiment analysis, and even code generation, making them powerful tools for numerous applications.

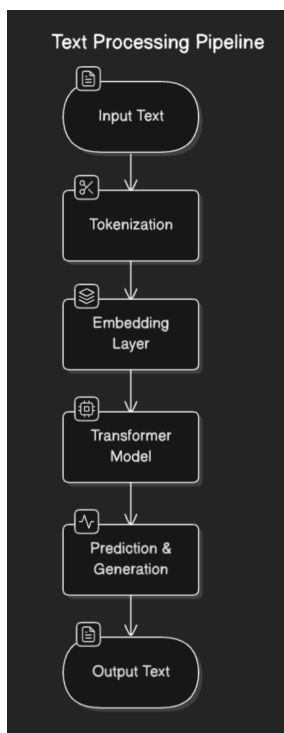


Figure 1: Text processing pipeline

While many organisations rely on commercial AI models like OpenAI's GPT or Google's Gemini, there are compelling reasons to build a custom LLM:

- ➔ Tailoring the model for industry-specific needs, such as legal, healthcare, or finance, ensuring better accuracy and relevance.
- ➔ Reducing reliance on expensive API calls from proprietary providers, leading to long-term cost savings.
- ➔ Enhancing data privacy and security by keeping sensitive information in-house rather than relying on third-party providers.
- ➔ Fine-tuning models for specialised tasks, improving performance over generic solutions.
- ➔ Maintaining independence from commercial AI providers and their pricing models or usage restrictions.

Open source models like BLOOM, GPT-Neo, and Llama provide flexible alternatives to proprietary models while requiring technical expertise for setup and maintenance. When deciding between open source and proprietary solutions, key differences must be considered (Table 1).

What you should consider before building your own LLM

Here's a detailed breakdown of crucial factors.